Internship – II

(UCS24P02L)
On

BillBuster – AI-Powered Financial Guard

At

LEARN-X ACADEMY GLOBAL

By

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Submitted to
DEPARTMENT OF COMPUTER SCIENCE W/S IN AI & ML

Under the guidance of

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SRM INSTITUTE OF SCIENCE & TECHNOLOGY

Ramapuram, Chennai.

FACULTY OF LIBERAL ARTS AND BUSINESS STUDIES

OCT / NOV 2026

SRM INSTITUTEOFSCIENCEANDTECHNOLOGY
Chennai Ramapuram

FACULTY OF LIBERAL ARTS AND BUSINESS STUDIES

Department of Computer Science w/s in ***AI & ML***

BONAFIDE CERTIFICATE

This is to certify that this Summer Internship - II (UCS24P02L) entitled ***" BillBuster – AI-Powered Financial Guard"*** is the bonafide work of ***"S Sana Fathima" (RA2431022020064)*** from ***B.Sc – AI & ML***, who carried out this summer internship under my supervision.

INTERNAL GUIDE ***HEAD OF THE DEPARTMENT***

Submitted for the University Practical Examination held at SRM Institute of Science and Technology, Ramapuram Campus on _____.

INTERNAL EXAMINER 1 ***INTERNAL EXAMINER 2***

OFFER LETTER

LEARN-X ACADEMY GLOBAL

Internship Offer Letter

Date: 25 May 2026

To,

Ms. Sana Fathima

Dear Ms. Sana Fathima,

We are pleased to offer you an internship position with Learn-X Academy Global under our Teaching & Training Internship Program, specializing in the Data Analytics domain. This letter outlines the terms of your internship.

1. Position: Intern – Data Analytics

2. Duration: Three (3) months, from 01 June 2026 to 31 August 2026.

3. Work Mode / Location: Virtual / Remote

4. Stipend: ₹1,500 per month

5. Responsibilities: You will assist in delivering training sessions, preparing learning materials, and supporting data analytics coursework as guided by your program mentor.

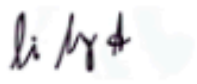
This offer is extended on the terms outlined above. On successful completion of the internship, a Certificate of Internship will be issued.

We look forward to having you on board.

Sincerely,

Asif Ahamed A

Founder & Director, Learn-X Academy Global



INTERNSHIP COMPLETION CERTIFICATE

LEARN-X ACADEMY GLOBAL

CERTIFICATE

This is to certify that Mr./Mrs./Ms.

SANA FATHIMA

has successfullycompletedtheTeaching &TrainingInternship Program at Learn-XAcademyGlobal,specializing intheDataAnalytics domain

for the period from 01 June 2026 to 31 August 2026,
with a duration of three (3) months.

ASIF AHAMED A
FOUNDER & DIRECTOR , LEARN-X ACADEMY GLOBAL

ABSTRACT

This internship provided practical exposure to the fields of Data Analytics and Data Science, with an emphasis on applying analytical concepts to real-world data. The primary objective of the internship was to develop practical skills in data collection, preprocessing, cleaning, analysis, visualization, and interpretation. During the internship, I gained hands-on experience with tools and technologies including Python, SQL, Microsoft Excel, and Power BI.

I worked with structured datasets to identify trends, detect inconsistencies, generate meaningful insights, and present analytical findings through interactive dashboards and visual reports. The experience also provided an opportunity to understand how analytical techniques can support efficient decision-making in practical business scenarios.

As part of the internship, I also developed a mini project titled "*BillBuster*," a web-based application designed to extract and analyze information from digital bills and receipts. The application uses Optical Character Recognition (OCR) to extract relevant details such as item names, prices, taxes, discounts, and total amounts. The extracted data is then cleaned and analyzed to identify useful patterns and potential anomalies, such as duplicate items, inconsistent discount calculations, and unusual values. The results are presented through simple visualizations and summarized insights to make the information easier to understand.

The internship enhanced my understanding of the complete data analysis workflow, from raw data collection to insight generation. It also improved my problem-solving, analytical thinking, data visualization, and technical skills. Overall, the experience provided a strong foundation for applying Data Analytics and Data Science techniques to practical, user-oriented applications while strengthening my ability to communicate data-driven findings effectively.

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1. INTRODUCTION

1.1 Bills, Receipts and Everyday Data

Every purchase produces a bill or receipt, andeach oneisasmall record of business data. A typical receipt contains the merchant name, thedate andtime,a list of items with quantities and prices, taxes, discounts, and a final total.Individuals,small businesses, and accounting teams handle a large number of such documents,and mostofthem are still stored as paper slips or as images and PDF files. Because this information isnotheld in a structured form, it is difficult to search, compare, or verify. The BillBuster projectdeveloped during this internship converts such receipts into structured data and checks that dataautomatically.

1.2 Role of Data Analytics in Expense Verification

Data analytics helps convert scattered records into information that can be checked and understood. When receipt data is captured in a structured format, calculations can be repeated automatically and patterns can be measured. Analytics can be applied to several areas of receipt and expense management, including:

- Receipt digitization
- Item and price extraction
- Total and tax verification
- Duplicate item detection
- Discount consistency checking
- Spending analysis by category
- Weekly and monthly spending trends
- Flagged and verified receipt tracking

The main advantage is that a task that would normally require careful manual reading can be handled consistently by a defined set of rules.

1.3 Importance of Automated Receipt Auditing

Errors on bills are common and are often not noticed. Aprice may be entered incorrectly, an item may be charged twice, a discount may be calculatedwrongly, or the printed total may not match the sum of the items. Checking these manually isslow and depends on the attention of the person reading the bill. An automated audit comparesthe printed total with the sum of the extracted items and highlights any difference. This givesthe user clear evidence about which receipt or item needs attention.

1.4 Need for Optical Character Recognition

Optical Character Recognition, commonly called OCR,is the technology that reads printed text from an image and converts it into machine-readabletext. For receipts, OCR is the first step, because the photograph of a bill contains no usable datauntil its characters are recognized. The quality of OCR depends on lighting, print clarity, andthe condition of the paper. BillBuster therefore treats OCR output as an input that must be validated instead of a result that can be trusted blindly.

1.5 Introduction to the BillBuster Application

BillBuster is a web-based application designed to extractand analyze information from digital bills and receipts. A user captures or uploads a receipt image, and the application extracts item names, prices, taxes, discounts, and total amounts. The extracted data is cleaned, checked against validation rules, and stored in a database. The results are presented through simple visualizations and summarized insights so thatthe information is easier to understand.

1.6 Overview of the Technologies Used

The application is built on Python and uses Pytesseract for OCR, regular expressions for entity extraction, a validation engine based on checksums and rules, SQLite as the relational store, and Plotly for the analytics dashboard. During the internship, further exposure was gained to SQL, Microsoft Excel, and Power BI, which supported the wider learning in data collection, cleaning, analysis, and visualization.

1.7 Application Concept

The application is designed around a simple user journey. The user scans a receipt, reviews the extracted items, and is told immediately whether the receipt is consistent. If a problem is found, the affected item is highlighted together with the expected value, and the user can correct it. Verified receipts are saved to a history and contribute to spending insights.

The application includes:

- A Scan screen for capturing or uploading a receipt
- A receipt detail screen showing items and mismatches
- An Insights screen with total spending, category split, and weekly spending
- A History screen with search and Flagged / Verified filters

1.8 Key Performance Indicators

Key indicators are measurable valuesthat summarize how well the system performs. In BillBuster the important indicators aretheaccuracy of the math rule validation, the precision of OCR under different conditions, theresponse time of database operations, and the time taken to complete a full audit. On the user side,indicators such as total spent, spending by category, and the number of flagged receipts give animmediate summary of the data.

1.9 Importance of Data Visualization

Numbers presented only as rows are difficultto interpret. Visualization makes comparisons, proportions, and trends easier to see.

For example:

- A donut chart shows the share of each spending category.
- A bar chart compares spending across weeks.
- A summary card displays the total amount spent.
- A list with status icons shows which receipts are flagged or verified.

In this project, visual elements are chosen so that a user can understand the result of an audit at a glance.

1.10 Interactive Application Environment

BillBuster is interactive instead of being a static report. The user can filter the receipt history by status, search for a receipt,select a month in the insights view, and open a receipt to see its individual items. This allowsthe user to explore the data according to their own needs.

1.11 Project Overview

The project follows a complete data analytics workflow. Receipt images are collected and passed through OCR. The recognized text is parsed into structured entities, and the entities are validated using mathematical rules. Valid records are stored in a relational database, and the stored data is presented through an analytics dashboard. The system was tested for accuracy, speed, and behavior under poor image quality.

2. PROJECT BACKGROUND AND BUSINESS NEED

2.1 Background of the Project

The BillBuster project was developedasa mini project during the internship in Data Analytics and Data Science. It demonstrates howdata extraction, cleaning, validation, and visualization can be combined into a single practicalapplication. The project focuses on transforming raw receipt images into reliable, structuredinformation.

2.2 Challenges in Manual Receipt Checking

Manual checking of receipts requires reading every line, adding the prices, and comparing the result with the printed total. Thisbecomes tiresome when many receipts are involved. Small errors such as a wrong price orarepeated item are easy to miss, especially when the receipt is long or the print is faint. Beforeautomation, a complete check of one receipt took around 120 seconds in the test comparison,which is not practical for regular use.

2.3 Item and Price Errors

An item on a bill may be chargedat a price different from the expected price. In such cases the total on the receipt no longer agrees with the individual lines. Identifying which item caused the difference requires comparing each line separately. BillBuster performs this comparison automatically and points to the exactitem that has caused the mismatch.

2.4 Total and Tax Mismatches

The printed total of a receipt shouldequal the sum of its items plus tax, minus any discount. When it does not, the receipt is inconsistent. Such mismatches may be caused by billing errors, incorrect discounts, or by characters that were read wrongly during OCR. The application reports the receipt total, the sum of items, and the difference between them so that the reason can be investigated.

2.5 Duplicate and Unusual Values

Receipts may contain duplicate items or values that look unusual compared with the rest of the bill. These may be genuine purchases or accidental entries. Detecting them manually is difficult, whereas a rule-based analysis can flag them consistently for the user to review.

2.6 Spending Tracking Difficulty

Individual receipts do not give a pictureof overall spending. A user who wants to know how much was spent on groceries, fuel, officesupplies, or dining in a month would need to collect all receipts and add them separately. Storing verified receipts in a database allows this summary to be produced automatically.

2.7 Image Quality Problems

Receipts are often photographedunder poor lighting, folded, or printed on thermal paper that fades over time. These conditionsreduce OCR accuracy. A practical system must be able to recognize when the extracted textisunreliable and warn the user instead of silently accepting incorrect data.

2.8 Need for Centralized Storage

Paper receipts and loose image files areeasily lost and cannot be searched. A central database gives a single place where receipts, items, and their status are stored, and where they can be retrieved quickly.

2.9 Need for Interactive Analysis

Users need to search, filter, and exploretheir receipts. A History view with status filters and a monthly Insights view removes the needto prepare separate reports each time.

2.10 Business Need Addressed by the Project

The project addresses the need for a fast and simple method of converting receipts into checked, structured data. The major areas addressed are:

- Receipt digitization
- Automated math validation
- Anomaly and mismatch detection
- Structured storage
- Category-wise spending analysis
- Weekly spending analysis
- Flagged and verified receipt tracking

3. PROJECT AIM AND OBJECTIVES

3.1 Project Aim

The primary aim ofthe project is to develop a web-based application that extracts information from digital bills and receipts, verifies it, and presents useful insights. The project combines image capture, OCR, data parsing, validation, database storage, and visualization.

3.2 Data Collection and Understanding Objective

The first objective is to understand the structure of a receipt and the fields it contains. Knowing which fields are present, such as merchant, date, items, prices, tax, and total, is necessary before designing the extraction logic.

3.3 Data Extraction Objective

The project aims to extract text fromreceipt images using OCR and to convert that text into structured entities. Item names, quantities, prices, taxes, discounts, and totals should be identified accurately.

3.4 Data Cleaning Objective

The extracted data is expected to contain noise such as misread characters, extra symbols, and inconsistent formatting. The objective is to clean this data so that it can be used reliably in calculations.

3.5 Validation Objective

A major objective is to verifyeach receipt using checksums and rules. The sum of the items must agree with the receipt total, and any difference must be reported clearly along with the item responsible.

3.6 Anomaly Detection Objective

The project aims to detect usefulpatterns and potential anomalies such as duplicate items, inconsistent discount calculations,and unusual values.

3.7 Data Storage Objective

Validated receipts are to be storedin a relational database so that they can be retrieved, filtered, and analyzed later. The databasemust respond quickly to create, read, update, and delete operations.

3.8 Spending Analysis Objective

The application should summarize stored receipts into total spending, spending by category, and weekly spending, so that the user gets aclear picture of expenses.

3.9 Interactive Reporting Objective

The project aims to provide an interactiveinterface with search, status filters, and month selection, rather than a static output.

3.10 User Interface Design Objective

The interface should be clear and easy to use.Important actions such as scanning and fixing a flagged item should be prominent, and statusindicators should make it easy to see which receipts need attention.

3.11 Performance Objective

The system should complete thefull cycle from scanning to saving quickly, and it should remain reliable when the image quality is poor.

4. SCOPE AND LIMITATIONS

4.1 Overall Scope

The project covers thedevelopment of a receipt analysis application. The scope includes image capture, OCR, entityextraction, validation, database storage, and visual analysis of spending. The work concentrateson the fields typically found on retail receipts.

4.2 Image Capture Scope

Receipts can be captured through the camera or uploaded from the device. The Scan screen guides the user to align the receipt within a frame and indicates when the lighting is suitable.

4.3 OCR and Extraction Scope

The application reads printed text fromreceipt images using OCR and extracts entities such as merchant name, date, itemnames, prices, taxes, discounts, and totals using pattern matching.

4.4 Validation Scope

Validation covers the comparison between the receipt total and the sum of the items, together with rule-based checks on quantities and prices. Receipts that fail a check are marked as flagged.

4.5 Storage Scope

The application storesvalidated receipt records in a relational SQLite database. The stored data supports the history view and the spending insights.

4.6 Spending Analysis Scope

The insights include the total amount spent in a selected month, the distribution of spending across categories, and the weeklyspending pattern.

4.7 Receipt History Scope

The history view lists stored receipts with merchant, date, amount, and status. It supports searching and filtering by All, Flagged, and Verified receipts.

4.8 Limitations

The project has certain limitations that should be noted:

- OCR accuracy fallswhenreceipts are folded, faded, or photographed in low light.
- The extraction rulesaredesigned for common receipt layouts and may need adjustment for unusualformats.
- Testing was carriedoutonalimited set of synthetic and real receipts.
- The application reliesontheprinted values and cannot confirm whether the merchant's price iscorrectinthe real world.
- Category assignmentdependson the merchant and item information available.

4.9 Visualization Scope

The project includes the followingvisual elements:

- Summary cards
- Donut chart
- Bar chart
- Receipt lists
- Status indicators
- Filters and search

5. TECHNOLOGY AND DEVELOPMENT ENVIRONMENT

5.1 Introduction

BillBuster was developed using a Python-based technology stack. Each stage of the workflow, from image capture to dashboard, is handled by a specific component. The tools were selected so that the complete data analytics process could be implemented in a simple and reliable manner.

5.2 Python

Python is themain programming language of the project. It is used to connect the OCR engine, parsing logic,validation rules, database, and dashboard into a single workflow. Python was also used during the internship for datapreprocessing and analysis tasks.

5.3 Pytesseract (OCR Engine)

Pytesseract is the Python interface tothe Tesseract OCR engine. It converts the receipt image into raw text. The quality of its outputdepends on the input image, which is why the application also checks the image conditions andvalidates the extracted data.

5.4 Regular Expression Parser

Regular expressions are used to recognize patterns in the OCR text. The parser identifies entities such as prices, quantities, item names, dates, taxes, and totals, and converts them into a structured JSON format.

5.5 Validation Engine

The validation engine applies checksumsandrules to the parsed data. It calculates the sum of item amounts, compares it with the receipttotal, and reports any mismatch. Receipts that pass are marked as valid, and those that failareflagged.

5.6 SQLite Database and SQL

SQLite is used as the relational storeforreceipts and items. SQL statements are used to insert, retrieve, update, and delete records.SQLitewas chosen because it is lightweight, requires no separate server, and performs quicklyforthisscale of data.

5.7 Plotly Dashboard

Plotly is used to build the analytics user interface. It produces the interactive charts for category-wise spending and weekly spending.

5.8 Microsoft Excel and Power BI

Microsoft Excel and Power BI were usedduring the internship for data cleaning, analysis, and dashboard building on structured datasets. This experience influenced the design of the spending insights in BillBuster.

5.9 Development Environment

The development environment mainlyconsisted of:

- Language: Python
- OCR Engine: Pytesseract
- Data Extraction: Regular Expressions
- Validation: Checksums and Rules
- Database: SQLite
- Visualization: Plotly

The components were combined into a single pipeline, as shown in Chapter 9.

6. RECEIPT DATASET

6.1 Dataset Overview

The data used in BillBuster consists ofbill and receipt images and the structured records extracted from them. Unlike a fixed table,the dataset is created by the application itself as each receipt is scanned. The receipts used fortesting included both synthetic and real examples so that the system could be evaluated undercontrolled and realistic conditions.

6.2 Receipt Header Information

The header of a receipt provides the merchant name and the date and time of the transaction. In the application, receipts from merchants such as Trader Joe's, Shell Gas, Office Depot, Starbucks, and CVS Pharmacy are displayed in the history. This information is used for identification, searching,andcategory assignment.

6.3 Item Information

Item information forms thecore of the data. Each item has a name, a quantity, a unit price, and a line amount. For example,items on a grocery receipt include Organic Bananas, Almond Milk, Sourdough Bread, and SeaSalt, each displayed with its quantity and price.

6.4 Tax and Discount Information

Taxes and discounts are stored separatelyfrom item amounts so that they can be included correctly in the total calculation. Inconsistent discount calculations are one of the anomalies that the application looks for.

6.5 Total Information

Each receipt has a printedtotal. The application also calculates the sum of the extracted items. The comparison betweenthe twoforms the basis of the mismatch check.

6.6 Category Information

Spending is classified into categories to support analysis. The categories shown in the insights view are:

- Groceries
- Gas
- Office

- Dining

The category information isused to build the spending by category chart.

6.7 Status Information

Each stored receipt has a status. A receipt is marked Verified when it passes the validation, and Flagged when an issue such as a mismatch is detected. These values are used in the History filters and in the count of flagged receipts.

6.8 Test Dataset

A set of 50 syntheticand real receipts was used to test the math rule validation. Separate image samples were usedto test OCR under optimal conditions (flat, high-contrast print) and degraded conditions(low light, folded, faded print).

7. DATA PREPARATION AND TRANSFORMATION

7.1 Importance of Data Preparation

Raw OCR output is rarely perfect. Extra symbols, wrongly read digits, and broken lines can produce incorrect calculations. Data preparation is therefore an important stage that makes the extracted data suitable for reliable analysis.

7.2 Image Capture and Input

The process begins with image capture. The user either photographs the receipt with the camera or uploads an existing image. The Scan screen helps the user align the receipt within a frame and shows a message when lighting is good, so that the image supplied to the OCR engine is as clear as possible.

7.3 OCR Text Extraction

The image is passed to the OCRengine, which returns raw text. This text contains all the visible characters of the receipt, including header lines, item lines, subtotals, taxes, totals, and barcodes.

7.4 Text Cleaning

The raw text is cleaned by removing unwanted symbols and extra spaces and by correcting common recognition errors. Lines that do not contain useful information, such as promotional messages, are ignored during parsing.

7.5 Entity Extraction Using Regex

Regular expressions identify item names,quantities, prices, taxes, and totals from the cleaned text. The extracted entities are convertedinto a JSON structure, which is passed to the next stage.

7.6 Data Type Validation

Prices, quantities, taxes, and totals must be numerical values, and dates must be in a valid date format. Correct data types prevent calculation errors and allow the values to be stored correctly in the database.

7.7 Standardization and Duplicate Checking

Merchant names, categories, and status values followconsistent naming so that similar records are grouped together. Duplicate items are examinedduring preparation because they may represent either a genuinerepeated purchase or an accidental repetition.

7.8 Data Validation

After parsing, the validation engine applies checksums and rules. Only receipts that pass are stored as valid records, while those with issues are flagged for review. The behavior of this stage was tested with 50 receipts, as described in Chapter 10.

8. DATA MODELING AND MEASURES

8.1 Introduction to Data Modeling

Data modeling involves organizing datasothat it can be stored and analyzed efficiently. In BillBuster, a relational model is used, inwhich receipts and their items are stored in linked tables in SQLite.

8.2 Field Classification

The fields of the model canbe classified as follows.

Numerical Fields

- Quantity
- Unit price
- Item amount
- Tax
- Discount
- Receipt total

Categorical Fields

- Merchant
- Category
- Status (Flagged / Verified)

Date Fields

- Receipt date
- Receipt time
- Month
- Week

This classification determineshow each field is stored and used in calculations and visuals.

8.3 Relational Structure

Each receipt is stored as a record with its merchant, date, total, category, and status. Each item is stored as a separate record linked to its receipt. This structure allows a receipt to have any number of items and allows the items to be added again when the receipt total is verified.

8.4 Validation Measures

The main validation measuresare calculated for every receipt:

- Receipt total: the amount printed on the receipt
- Sum of items: the total of all extracted item amounts
- Mismatch: the difference between the receipt total and the sum of items

A receipt with a mismatch different from zero is marked as flagged.

8.5 Worked Example of Validation

The Trader Joe's receipt in theapplicationshows a receipt total of $23.76 and a sum of items of $33.76, giving a mismatchof $10.00.The validation engine traced this to the item Sourdough Bread, which wasrecorded at $14.99 whereas the expected price was $4.99. The difference between these pricesexplains the whole mismatch.

8.6 Spending Measures

The insights are producedbyaggregating stored receipts:

- Total Spent = sumofreceipt totals in the selected month
- Spending by Category=sum of totals grouped by category
- Weekly Spending=sumof totals grouped by week
- Flagged Receipts=countof receipts with a flagged status

For September, the applicationshows a total spent of $1,284.52.

8.7 Query Performance

SQL queries are used foralldatabase operations. In testing, 1,000 concurrent create, read, update, and delete querieswereexecuted with each query completing in under 8 ms, which shows that the modelissuitablefor interactive use.

8.8 Filter Context

Selections made by theuser change the data displayed. Choosing a month updates the total spent, the category split,and the weekly chart. Choosing the Flagged or Verified filter in the history updates the list ofreceipts. This behavior makes the application interactive.

9. APPLICATION DEVELOPMENT

9.1 Application Planning

Before development,theapplicationwas divided into functional stages and screens. The workflow was plannedasasix-stagepipeline, and the user interface was planned as three main screens: Scan, History,andInsights,with a receipt detail screen for reviewing individual receipts. This separationpreventstoomuch information from appearing on a single screen.

9.2 Execution Pipeline

The pipeline consists of sixstages.Image Capture receives the receipt through the camera or upload. The OCR Engine,usingPytesseract, converts the image into text. The Regex Parser extracts entities from thetextandproduces JSON. The Validation Engine applies checksums and rules to the entities.Validrecords are saved in the SQLite database, and the Plotly Dashboard displays analyticsfromthestored data.

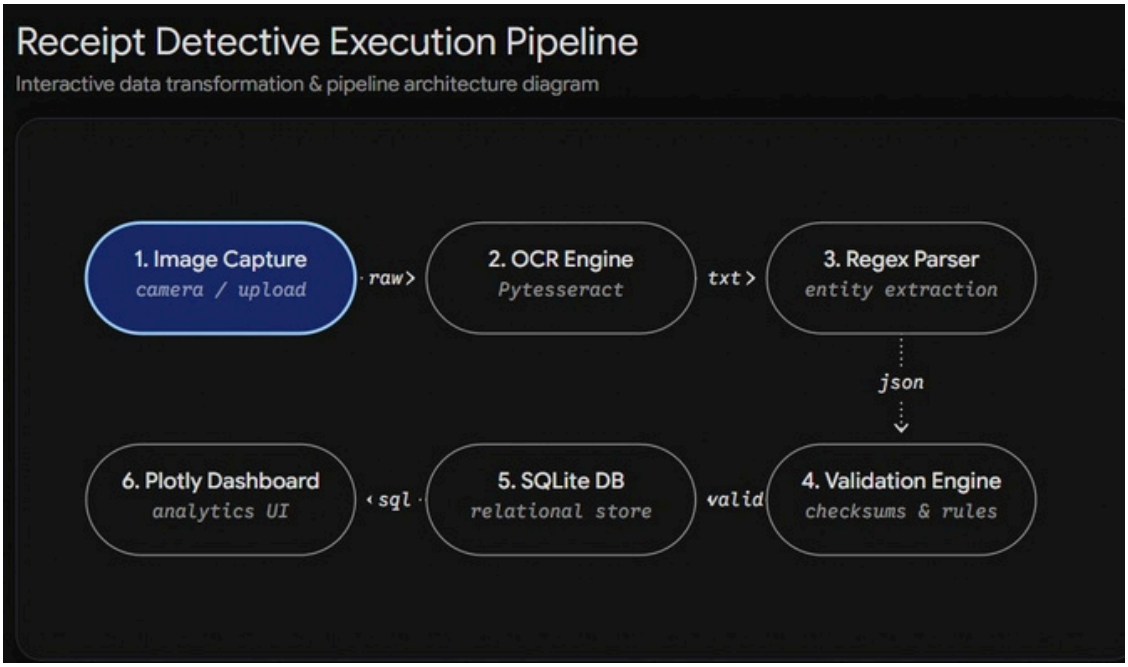


Figure:1: Receipt Execution Pipeline

9.3 Scan Screen

The Scan screen isthe entry point of the application. It shows the camera view with a frame for aligning the receipt, along with a message indicating that the lighting is good. A large scan button, an image upload button, and a flash button are provided. Recent scans are shown at the bottom, and navigation tabs give access to Scan, History, and Insights.

9.4 Receipt Detail and Flagged Item Screen

After scanning, the receipt detail screen lists the merchant, the date and time, and the items with their quantities and prices. A message shows the number of issues detected. The flagged item is highlighted, and the expected price is displayed beneath it. A summary section shows the receipt total, the sum of items, and the mismatch. A Fix flagged item button allows the user to correct the item.

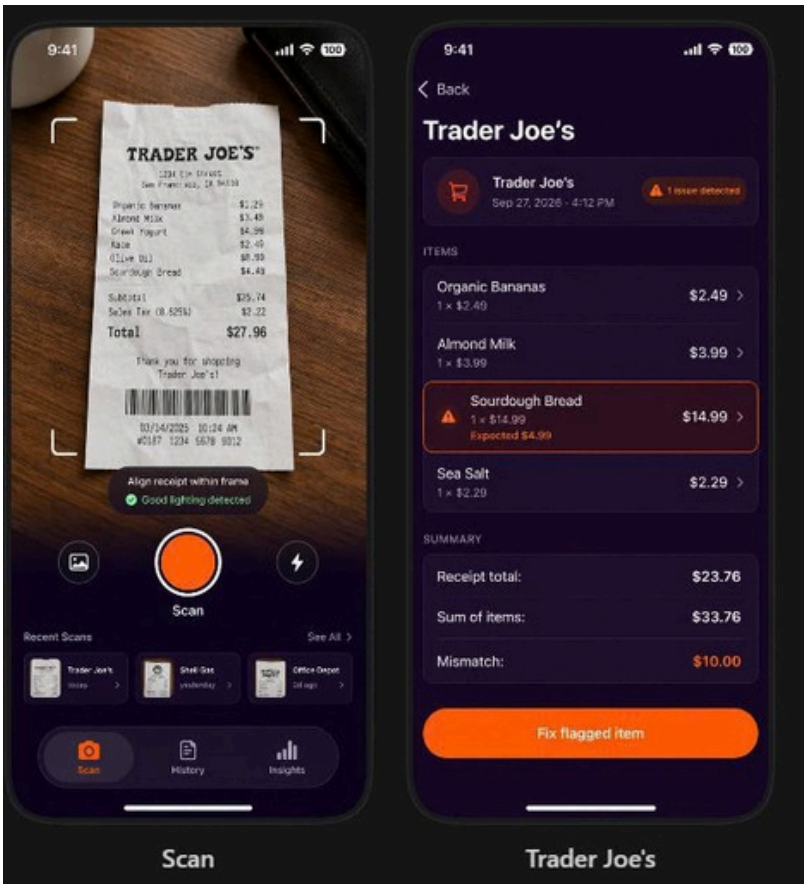


Figure:2: Scan and Receipt Detail Screens

9.5 Insights Screen

The Insights screen summarizes spending for the selected month. A card at the top shows the total spent. A donut chartshows spending by category, and a bar chart shows weekly spending. A banner at the bottomshows how many receipts were flagged during the month.

9.6 History Screen

The History screen liststhestored receipts with the merchant logo, date, amount, and status icon. A search box allowsthe user to find a receipt, and the All, Flagged, and Verified tabs filter the list. Flagged receipts are marked with a warning icon and verified receipts with a tick.

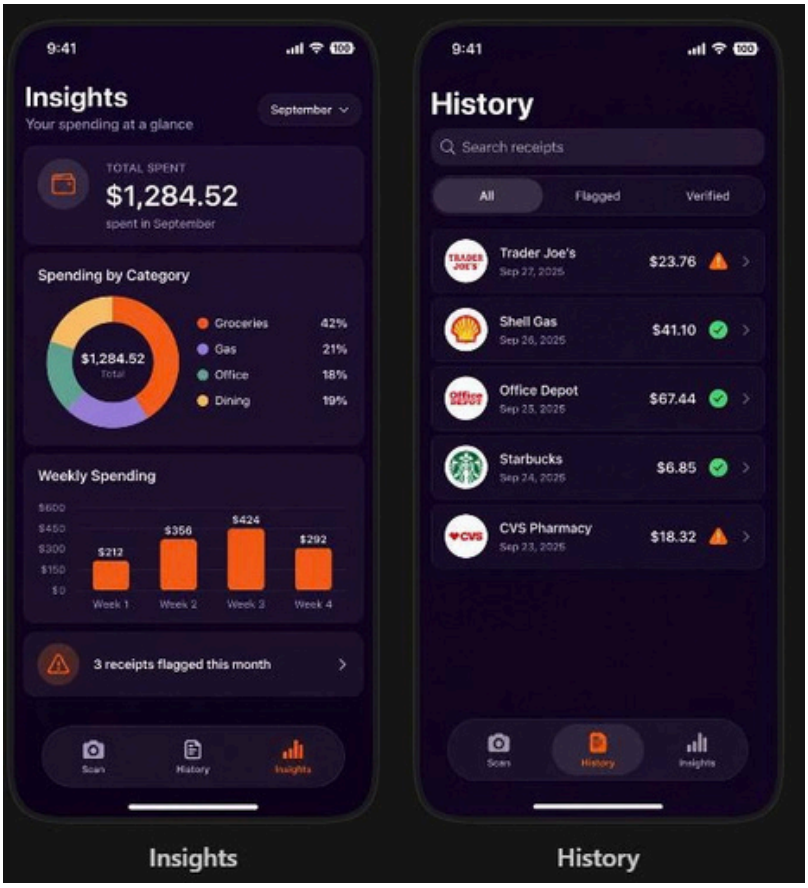


Figure:3: Insights and History Screens

9.7 Chart Selection

Different visualizationswere selected for different purposes. A donut chart shows the proportion of spendingineach category. A column chart shows the change in spending from week to week. Cards present summary values, and lists present detailed receipt information.

9.8 Design Considerations

A dark interface with an orangeaccent color is used so that important actions and warnings stand out. Flagged items are highlighted using a warning color, and the primary action on each screen is placed prominently for easy access.

10. ANALYSIS AND BUSINESS INSIGHTS

10.1 Overview

The completed application provides several analytical views. The following observations are based on the valuesdisplayed in the application and on the results of the test suite.

10.2 Test Results

The application wasevaluated with five test suites, covering validation accuracy, database speed, OCR precisionunder two conditions, and end-to-end speed. The results are summarized below.

Test Suite	*Test Scope*	*Result*	*Status*
Math Rule Validation	50 synthetic / real receipts	100% accuracy	Passed (zero discrepancies missed)
Database Latency	1,000 concurrent CRUD queries	< 8 ms execution	Passed (high throughput)
Optimal OCR Ingestion	Flat, high-contrast print	94.2% precision	Passed (clear text read)
Degraded OCR Ingestion	Low light, folded, faded print	76.5% precision	Passed (triggers audit flag)
End-to-End Audit Speed	Full scanning to save cycle	15 s average	Passed (reduced from 120 s)

Table 10.1 Test Suite Results

10.3 Validation Accuracy

The math rule validation achieved 100% accuracy on the 50 test receipts, and no discrepancies were missed. This showsthat the checksum approach is dependable for detecting inconsistencies between itemamounts and receipt totals.

10.4 OCR Performance

OCR precision was 94.2%forflat, high-contrast prints and 76.5% for low-light, folded, or faded prints. The differenceshows the effect of image quality. The lower precision in degraded conditions triggers an auditflag, so that unreliable results are brought to the attention of the user instead of being acceptedsilently.

10.5 Processing Speed

The full cycle from scanning to saving takes an average of 15 seconds, compared with 120 seconds for manual checking. This is a reduction of 105 seconds per receipt, or about 87.5%. Database operations completed in under 8 ms, so storage was not a limiting factor.

10.6 Flagged Receipt Analysis

In the History view, five recentreceipts are displayed. Trader Joe's ($23.76) and CVS Pharmacy ($18.32) are flagged,while Shell Gas ($41.10), Office Depot ($67.44), and Starbucks ($6.85) are verified.TheInsights view reports that 3 receipts were flagged during the month. This allows a usertoconcentrate on the receipts that need attention.

10.7 Spending Analysis

The application shows a totalspending of $1,284.52 for September. The distribution across categories is:

- Groceries: 42%
- Gas: 21%
- Dining: 19%
- Office: 18%

Groceries form the largest categoryandaccount for more than any other. Gas is second, and dining and office spending are closetoeach other.

10.8 Weekly Spending Pattern

The weekly spending values are:

Week 1: $212
Week 2: $356
Week 3: $424
Week 4: $292

Spending was highest in Week 3 and lowest in Week 1. This helps the user notice changes in spending during the month.

10.9 Key Findings

The main findings fromthe project are:

- Automated validation detected all discrepancies in the test receipts.

- Image quality is the main factor affecting OCR precision.
- Flagging low-precision results protects the user from incorrect data.
- Automation reduced audit time from 120 seconds to 15 seconds.
- Stored receipts provide useful summaries of spending by category and by week.

11. PROJECT OUTCOMES AND LEARNING

11.1 Major Project Outcome

The main outcome of the projectisthe successful development of BillBuster, a web-based application that extracts, validates,stores, and analyzes receipt data. It combines OCR, parsing, validation, database storage, andvisualization into one pipeline and meets its performance targets in testing.

11.2 Improved Data Analysis Skills

The project provided practical experience inworking with structured and semi-structured data. The development process requiredunderstanding the meaning of fields, identifying validation rules, detecting inconsistencies, andchoosing suitable ways to present results.

11.3 Technical Skills Gained

The internship and the project provided practical knowledge of:

- Python programming
- Optical Character Recognition using Pytesseract
- Regular expressions
- SQL and SQLite
- Data cleaning and preprocessing
- Data validation techniques
- Data visualization using Plotly
- Microsoft Excel
- Power BI
- Interactive report and interface design

11.4 Data Cleaning Learning

The project showed that real-worlddata is rarely clean. OCR errors, inconsistent formatting, and missing values must be handledbefore any analysis. The importance of validating data at each stage became clear duringdevelopment.

11.5 Validation Learning

Simple rules such as comparing the sum of items with the total can be very effective. The project also showed the value of reporting the expected value and the exact item causing the mismatch, since this makes the result easy for a user to act on.

11.6 Data Visualization Learning

Different types of information require different visualization techniques. For example:

- Donut charts were used for proportions.
- Bar charts were used for weekly comparisons.
- Cards were used for summary values.
- Lists with status icons were used for detailedrecords.

11.7 Testing and Performance Learning

Testing under both good and poor conditions gavearealistic view of system performance. It highlighted how the quality of input affects output and why the system should signal uncertainty.

11.8 Communication of Findings

The project improved the ability tocommunicate data-driven findings. Presenting the mismatch, the expected price, and theresulting summary in a clear way showed how analysis can be made understandable for non-technical users.

12. CONCLUSION AND FUTURE ENHANCEMENT

BillBuster, developed as a mini project during the Data Analytics and Data Science internship, provides an efficient and practical solution for analyzing bills and receipts. The application combines image capture, OCR using Pytesseract, regex-based entity extraction, rule-based validation, SQLite storage, and a Plotly dashboard into a single workflow. By using this pipeline, unstructured receipt images are transformed into structured data that can be verified and analyzed.

The project helps identify errors such as item price mismatches, total discrepancies, duplicate items, and unusual values. In testing, the math rule validation reached 100% accuracy on 50 receipts, database operations completed in under 8 ms, and OCR precision reached 94.2% for clear prints and 76.5% for degraded prints, with the lower value triggering an audit flag. The average time for a complete audit was reduced from 120 seconds to 15 seconds. The Scan, Receipt Detail, Insights, and History screens make the results simple to understand and act upon. This project also gave practical experience in data cleaning, validation, analysis, visualization, and application development.

The project can be further improved in the future. OCR accuracy can be increased by adding image preprocessing such as noise removal, deskewing, and contrast enhancement, or by using deep learning based OCR models. Machine learning models can be introduced for automatic category prediction and for detecting unusual spending. Support for multiple languages and currencies, and for different receipt layouts, would broaden the use of the application.

Additional features such as cloud storage, user accounts, budget alerts, export of reports to Excel or PDF, integration with accounting tools, and a mobile application can make the system more useful. Automatic reminders for flagged receipts, warranty tracking, and integration with payment apps are further possibilities. These enhancements would improve the accuracy, usability, and scalability of the application. Overall, the project demonstrates how data analytics and OCR can be used to convert receipts into reliable insights and support better financial awareness and decision-making.